



Afformed Full Stack Assessment Preparation Checklist

Backend Fundamentals

HTTP Basics

- Client-Server Architecture
- Request & Response Lifecycle
- HTTP Status Codes
- REST Architecture

HTTP Methods

- GET
- POST
- PUT
- PATCH
- DELETE

Express.js

- Creating Express Server
- Routes
- Middleware
- Route Handlers
- Error Handling

Request Data Handling

- req.body
- req.params
- req.query
- Headers

API Design

- REST APIs
- Route Naming Conventions
- JSON Response Structure
- Response Status Codes

API Integration

Axios

- axios.get()
- axios.post()
- Request Configuration
- Timeout Handling

External API Consumption



- Calling Third-Party APIs
- Handling API Responses
- Handling API Failures

Multiple API Calls

- Promise.all()
- Promise.allSettled()
- Async/Await
- Parallel API Execution

API Aggregation

- Collecting Data from Multiple Sources
- Merging Responses
- Data Transformation

JavaScript Concepts

Modern JavaScript

- let / const
- Arrow Functions
- Destructuring
- Spread Operator
- Template Literals

Arrays

- map()
- filter()
- find()
- reduce()
- sort()
- flat()
- flatMap()

Objects

- Dynamic Property Access
- Object Manipulation
- Nested Objects

Error Handling

- try-catch
- Custom Error Responses
- API Error Handling



Data Structures & Algorithms

Arrays

- Traversal
- Merging Arrays
- Removing Duplicates
- Sorting
- Search

Strings

- Prefix Matching
- Unique Prefix Problems
- String Comparison

Hash-Based Structures

- Set
- Map

Time Complexity

- $O(n)$
- $O(n \log n)$
- Basic Complexity Analysis

Backend Engineering Concepts

Filtering

Examples:

- minPrice
- maxPrice
- category

Sorting

Examples:

- sort=price
- sort=rating
- asc / desc

Pagination

- page
- limit
- offset concept
- slice()

Search

- Keyword Search



- Query-Based Filtering

Data Aggregation

- Combining Multiple Vendor Responses
- Flattening Nested Arrays

Frontend Fundamentals

React Basics

- Components
- JSX
- Props
- Event Handling

React Hooks

- useState
- useEffect

Component Design

- Reusable Components
- Parent-Child Communication

API Integration

- Axios in React
- Fetching Data
- Loading Data on Page Load

State Management

- Managing API Responses
- Updating UI Dynamically

Frontend Features

Product Listing

- Display Data from APIs

Filtering

- Filter by Price
- Filter by Category

Sorting

- Sort by Price
- Sort by Rating

Pagination

- Previous / Next
- Page Numbers



Forms

- Input Fields
- Dropdowns
- Controlled Components

Full Stack Concepts

Backend + Frontend Integration

- React consuming Express APIs
- Handling JSON Responses
- Dynamic Rendering

CORS

- What is CORS?
- Why CORS Happens?
- Using cors Middleware

Folder Structure

Backend:

- Routes
- Controllers
- Services

Frontend:

- Components
- Pages
- Services

Git & GitHub

Git Basics

- `git init`
- `git add`
- `git commit`
- `git push`

GitHub Submission

- Repository Creation
- Proper Folder Structure
- Meaningful Commits



Most Important Topics (High Priority)

1. Express.js
2. REST APIs
3. req.body / req.params / req.query
4. Axios
5. Async/Await
6. Promise.all()
7. Promise.allSettled()
8. Error Handling
9. Arrays & Strings
10. Set & Map
11. Sorting
12. Filtering
13. Pagination
14. API Aggregation
15. React useState
16. React useEffect
17. API Integration in React
18. CORS
19. GitHub Workflow
20. Clean Folder Structure